Rui Yang

yangrui0122@foxmail.com | github.com/RuiYang0122 | ruiyang0122.github.io

Education

B.Eng. in Electronic Science and Technology **Sep 2023 – Jun 2027 (Expected)**
Shandong University, School of Information Science and Engineering, Chongxin College Overall GPA: 86.8/100

Internship Experience

Insta360 Research **Jul 2026 – Present**
Deep Learning Algorithm Intern

- **AIGC for controllable generation and editing:** developed controllable music generation methods based on Stable Audio Open 3, training adapters with binary MIDI sequences as temporal conditioning signals to enable control and editing of generated music; explored efficient few-step generation based on Flow Matching and multimodal reference-conditioned audio generation.
- **On-device vision algorithms (independently responsible):** trained object detection and keypoint (Bbox & Pose) models for panoramic UAVs; reduced on-device inference cost through convolution-operator optimization and model parameter compression; handled corner cases via AIGC-based data augmentation; designed a cross-frame tracking strategy for target association and state maintenance.

Shootz.tech × Central South University **Jan 2026 – Jul 2026**
Vision Algorithm Intern & Research Assistant Advisor: Prof. Shaokai Wang

- **Player Re-identification (ReID):** fine-tuned a Qwen3-VL-Embedding model with contrastive learning on text-semantic data, analyzed embeddings through dimensionality-reduction visualization, and optimized K-Means clustering to automatically associate player identities across clips.
- **Player Tracking:** trained detectors on 70,000+ basketball images and combined them with a cross-frame temporal state machine for global player tracking and event association.
- **Court2Stats research framework:** proposed a multi-agent framework for player-centric understanding, event analysis and game statistics; built the C2S-Bench multi-view benchmark and BasketWiki knowledge base for long-video player-level analysis and commentary.

Publications

Benchmarking Multimodal Knowledge Conflict for Large Multimodal Models arXiv:2505.19509
Yifan Jia, Kailin Jiang, Yuyang Liang, Qihan Ren, Yi Xin, **Rui Yang**, et al. *AAAI 2026 (Oral)*

Beyond a Single Crop: Contextualized Embeddings for Player Re-Identification arXiv:2610.13259
Ge Luo, Changhuo Yang, Weisi Fan, **Rui Yang**, Shaokai Wang *Pre-Print*

Towards Capability-Aware Multimodal Knowledge Conflict Detection and Resolution
Yifan Jia, **Rui Yang**, Qi Fan, Xiaoying Tang *Under Review*

Benchmarking Cross-Modality Safety Consistency in Large Vision-Language Models
Yuntao Du, Yifan Jia, **Rui Yang**, Dongrui Liu, Qian Li, Mingcai Chen, Lizhen Cui *AAAI 2027 (Under Review)*

Competitions & Projects

20th National University Students Intelligent Car Race — 5G Remote Autonomous Driving
Team Leader / Vision & Control Algorithm Lead **First Prize, North China Division (Ranked 2nd)**

- Designed a fill-rate rectangle filter for stable zebra-crossing recognition during 6 m/s line following, and built the lane-following pipeline: CLAHE + Canny + Hough lane extraction, inverse perspective mapping, and nonlinear PID steering.

HoopCut — Open-Source Basketball Highlight Auto-Editing System github.com/RuiYang0122/HoopCut

- ★ 20 stars / 4 forks on GitHub; YOLO-based shot-event detection with geometric trajectory fitting, and an FFmpeg pipeline that cuts scoring highlights end-to-end from raw footage.